

Education

École Polytechnique Fédérale de Lausanne, Switzerland	since 2024
Bachelor's degree in Physics	
V. gimnazija, Zagreb, Croatia	2020 – 2024
Grammar school of scientific-mathematical orientation	

Awards and Competitions

Platinum trophy, Physics Cup – Online Physics Competition	2025
Absolute winners, team with Ilan Mihelja, a fellow high-school student	
Gold trophy, Physics Cup – Online Physics Competition	2024
Team with Ilan Mihelja, a fellow high-school student	
33 rd place, Physics Brawl - International Online Physics Competition	2024
Group placement in the open category among 230 teams	
Silver trophy, Physics Cup – Online Physics Competition	2023
Team with Ilan Mihelja, a fellow high-school student	
Bronze medal, International Physics Olympiad (IPhO)	2023
Individual placement	
Bronze medal, European Physics Olympiad (EuPhO)	2023
Individual placement	
Silver medal, European Olympiad of Experimental Sciences (EOES)	2022
Group placement	
National physics and chemistry competitions (Croatia)	2021 – 2024
Includes a 1 st place (2021), a 2 nd place (2022), and three 3 rd places (2022, 2023, 2024)	

Teaching and Engagement

Student Assistant, EPFL	2025 – 2026
PHYS-105 (Prof. Cécile Hébert) – Advanced Physics II: Thermodynamics.	
PHYS-108 (Prof. Christian G. Theiler) – General Physics II: Fluids and Electromagnetism.	
PHYS-106(a) (Prof. Nicholas Grandjean) – General Physics II: Thermodynamics.	
PHYS-101(f) and (k) (Prof. Frédéric Mila) – General Physics I: Mechanics.	

Assisted with weekly exercise sessions, including answering student questions, correcting assignments, preparing exercise solutions, and organising problem-solving sessions with students in lecture halls.

Personal Projects

Composition of <i>Treatise on Plasma Physics</i>	since 2025
Development of a complete and rigorous treatise covering the theoretical foundations of plasma physics, aiming to offer an exhaustive and in-depth resource where classic works remain too introductory. Available on GitHub .	

Skills and Experience

Programming languages and data analysis

Highly proficient in C++ and Python: experienced in object-oriented programming, managing complex projects; vastly experienced with numerical methods and data analysis techniques in physics.

Workflow, tools and typesetting

Advanced proficiency in $\text{\LaTeX}$, including vector graphics libraries TikZ, PGFPlots, and general complex document structuring.

Comfortable with Unix/Linux system interfaces, terminals (Bash), and (Neo)Vim.

Outreach and Communication

Problem Writer - Online Physics Olympiad (OPhO) Committee 2025

Authored theoretical problems for the OPhO 2025, a competition held annually in late August with over 1000 participating teams. OPhO problems typically require great creativity and solid knowledge of physics. Available on the [OPhO website](#).

Problem Writer - Online Physics Challenge Problems 2025 – 2026

Composed multiple challenge problems in various fields of physics with the goal of contributing to the official Physics Discord server. Available on [GitHub](#).

PH++ Advanced Physics Series 2024 – 2025

Voluntarily and individually designed series of advanced exercises aiming to introduce fellow physics students at EPFL to problems comparable to those typically found in physics olympiads. Available on [GitHub](#).

Advanced Chemistry and Physics Classes 2023 – 2025

Voluntarily taught advanced chemistry and physics courses at V. gimnazija. Annual courses, held outside school hours throughout the year, primarily designed to prepare students for national and international competitions.

Research Interests and Future Objectives

Ongoing theoretical aspirations: Magnetic stochasticity in confined plasmas, Collisional transport in magnetised plasmas, Functional analysis for PDEs.

Ongoing computational aspirations: General computational methods in plasma physics, Gyrokinetic turbulent models.

About me:

I am a dedicated physics student planning to pursue a Master's and a PhD in theoretical and computational plasma physics at EPFL. My core research aspiration is to help develop current theoretical notions in plasma physics from the perspective of functional analysis and work on computational methods for confined plasmas. I am actively seeking to contribute to research projects at the Swiss Plasma Center. Beyond research, I am deeply passionate about pedagogical outreach and intend to remain involved in teaching and mentoring throughout my academic career.

Languages, by order of proficiency

Croatian (mother-tongue), English (C2), French (C1), Romanian (intermediate)

Contact information

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